

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T42 · Circle Theorems

Questions

Student book · questions with writing space

■ 17 pages

FAMILY MAP

Circle Theorems

F01	Use radius, chord and tangent structure	3 Q 14 marks
F02	Write justified circle-angle chains	11 Q 39 marks
F03	Use same-segment and cyclic-quadrilateral theorems	1 Q 4 marks
F04	Use intersecting chord and secant products	1 Q 3 marks

Question 24

4MA1-2H Q21

MODULAR UNIT 2

5 marks

- 21 The diagram shows a square $ABCD$ and a circle.

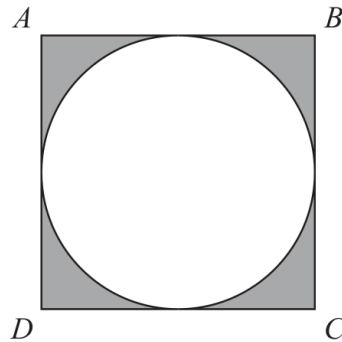


Diagram **NOT**
accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is 80 cm^2

Work out the length of AC

Give your answer correct to 3 significant figures.

..... cm

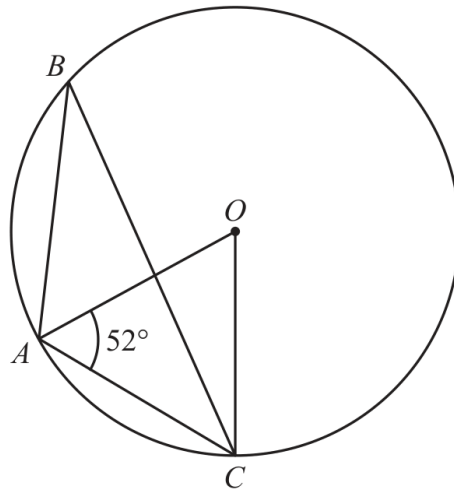
Question 11

4MA1-1H Q14 demand

MODULAR UNIT 2

3 marks

14

Diagram **NOT**
accurately drawn A , B and C are points on a circle, centre O Angle $OAC = 52^\circ$ Find the size of angle ABC

Give reasons for your working.

.....
○

Question 17

4MA1-1H Q22 demand

MODULAR UNIT 2

6 marks

- 22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

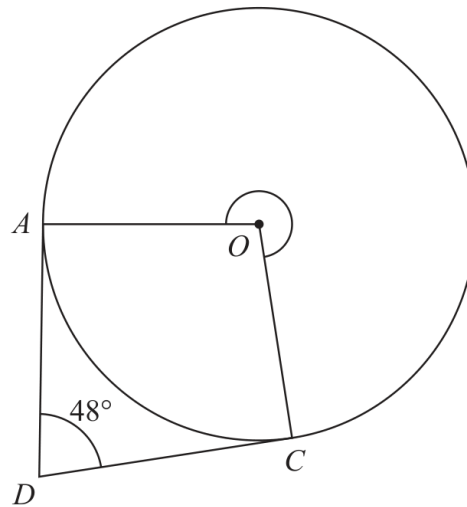
Question 21

4MA1-2H Q7

MODULAR UNIT 2

3 marks

7

Diagram **NOT**
accurately drawn

A and C are points on a circle, centre O

DA is the tangent to the circle at A and DC is the tangent to the circle at C

Angle $ADC = 48^\circ$

Work out the size of reflex angle AOC

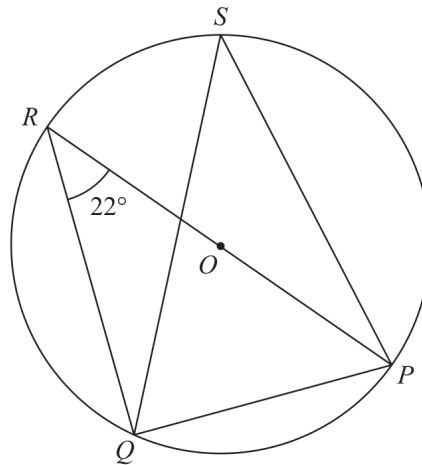
Question 25

4MA1-2H Q14

MODULAR UNIT 2

4 marks

14

Diagram **NOT**
accurately drawn

P, Q, R and S are points on a circle, centre O
 ROP is a diameter of the circle.
 Angle $PRQ = 22^\circ$

(a) (i) Find the size of angle RQP

.....
 (1)

(ii) Give a reason for your answer.

.....

 (1)

(b) (i) Find the size of angle PSQ

.....
 (1)

(ii) Give a reason for your answer.

Question 28

4MA1-2H Q16 M01, M02, M03, M04

MODULAR UNIT 2

5 marks

16 A, B, C and D are points on a circle, centre O

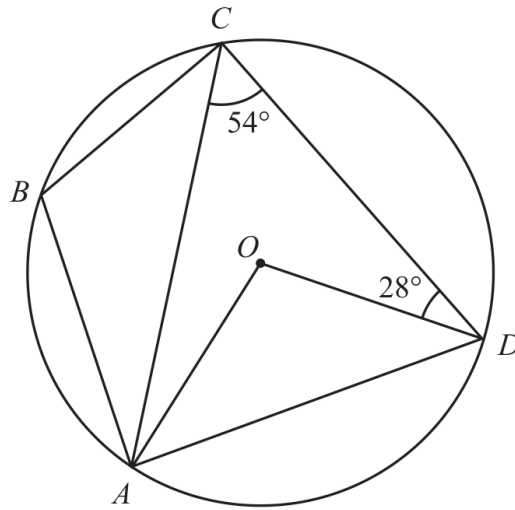


Diagram **NOT**
accurately drawn

- (a) (i) Work out the size of angle AOD

(1)

- (ii) Give a reason for your answer to part (a)(i)

(1)

- (b) Work out the size of angle CAO

(1)

- (c) Work out the size of angle ABC

Question 29

4MA1-2H Q16

MODULAR UNIT 2

4 marks

16 D, E, F and G are points on a circle, centre O

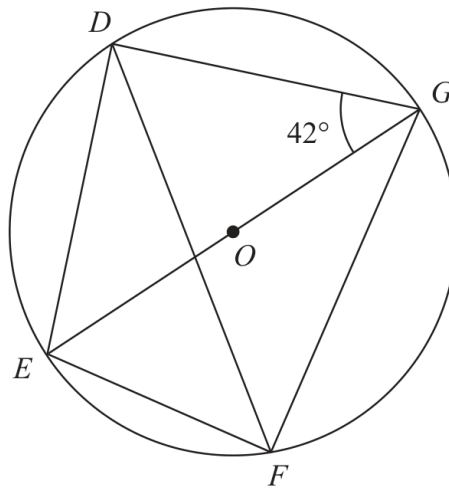


Diagram **NOT**
accurately drawn

EOG is a diameter of the circle.

Angle $EGD = 42^\circ$

Calculate the size of angle DFG

Give a reason for each stage of your working.

Angle $DFG = \dots\dots\dots^\circ$

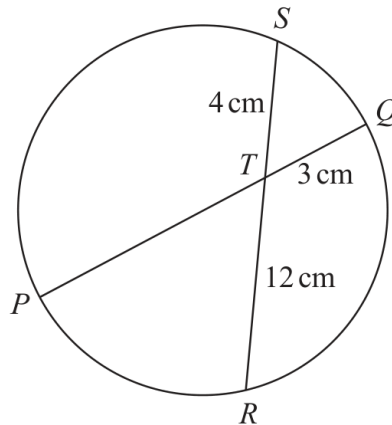
Question 31

4MA1-2H Q19

MODULAR UNIT 2

3 marks

19

Diagram **NOT**
accurately drawn PTQ is a diameter of a circle. RTS is a chord of the circle.

$TQ = 3 \text{ cm}$

$ST = 4 \text{ cm}$

$TR = 12 \text{ cm}$

Calculate the radius of the circle.

..... cm

A circle with center O . Points A , B , C , and D are on the circumference. Line segments AC and BD are drawn. Angle ABC is 28° and angle BDC is 32° .

Diagram **NOT**
accurately drawn

Find the size of angle BDC .
Give a reason for each stage of your working.

13 P, Q, R, S and T are points on a circle with centre O .

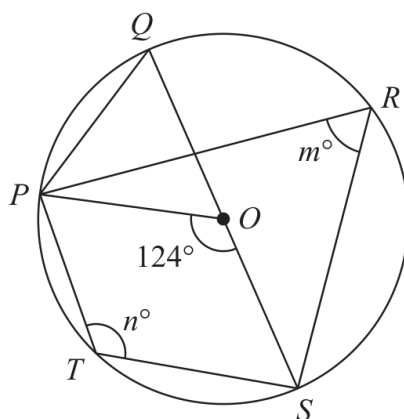


Diagram **NOT**
accurately drawn

QOS is a diameter of the circle.

angle $POS = 124^\circ$ angle $PRS = m^\circ$ angle $PTS = n^\circ$

(a) Find the value of

(i) m

(ii) n

(2)

(b) Find the size of angle QPO .

13 P, Q, R, S and T are points on a circle with centre O .

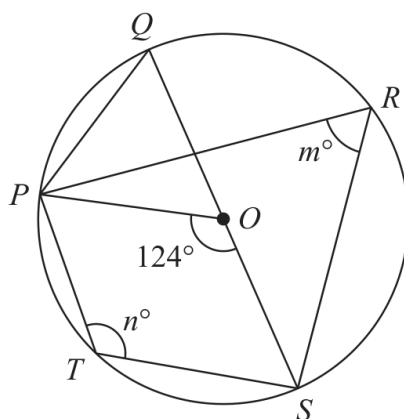


Diagram **NOT**
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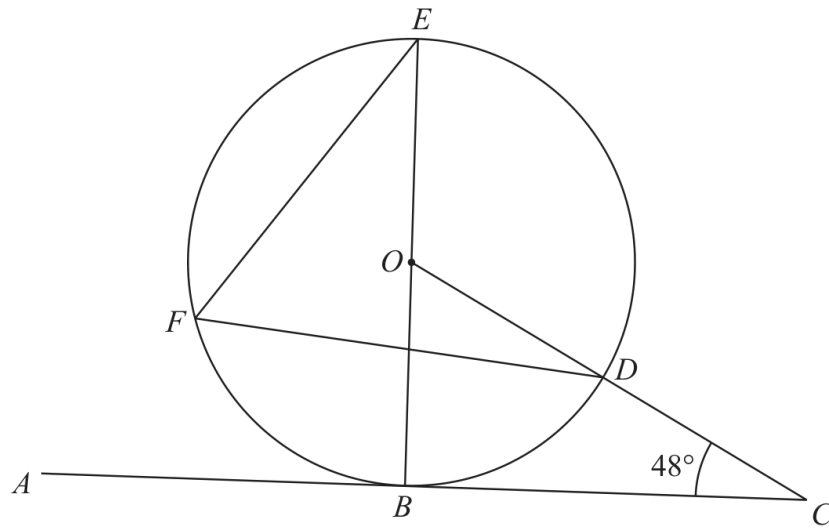
(2)

(b) Find the size of angle QPO .

-

Give a reason for each stage of your working.

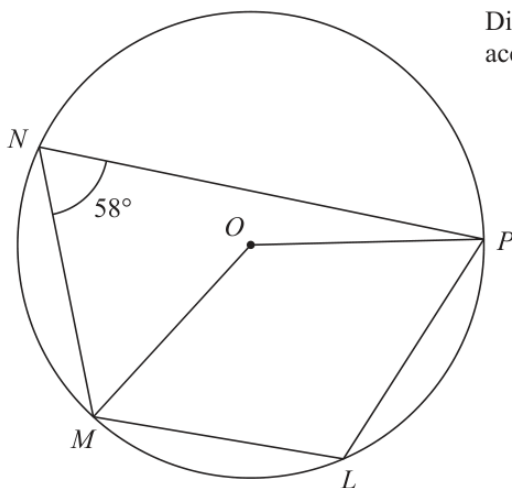
Diagram **NOT**
accurately drawn



Find the size of angle DFE .

16

Diagram **NOT**
accurately drawn



L, M, N and P are points on a circle, centre O

Angle $MNP = 58^\circ$

(a) (i) Find the size of angle MLP

o

(ii) Give a reason for your answer.

(2)

(b) Find the size of the reflex angle MOP

EBF is the tangent to the circle at B

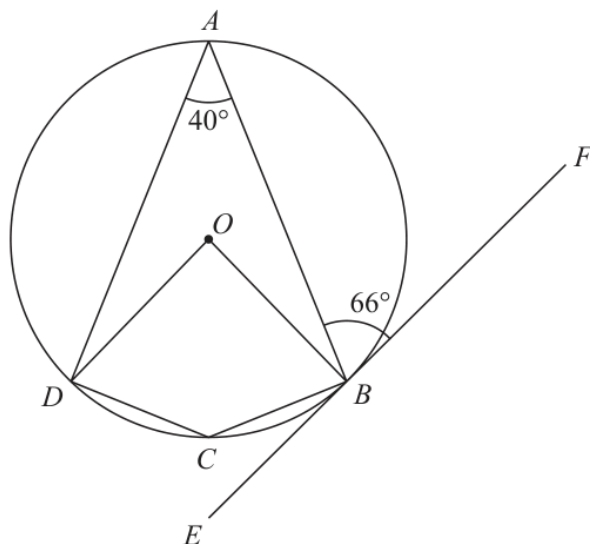


Diagram **NOT**
accurately drawn

- (a) (i) Work out the size of angle DCB

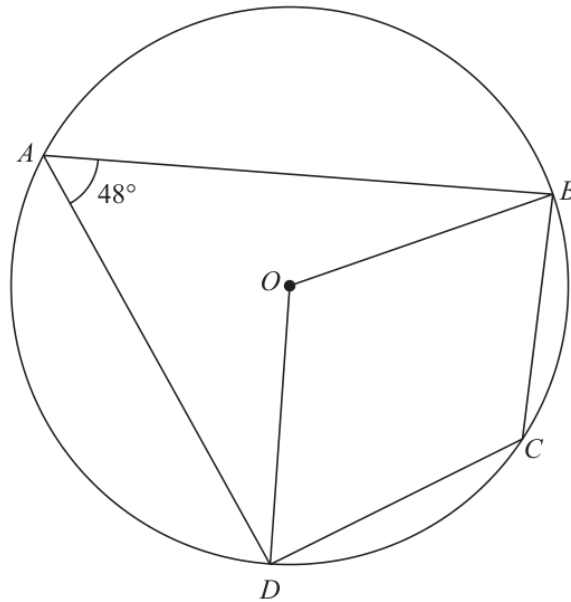
(1)

- (ii) Give a reason for your answer to (a)(i)

(1)

- (b) Work out the size of angle ADO

Diagram **NOT**
accurately drawn



(1)

(1)